

REVIEW

Are seed oils really bad for you?

REFERENCES	STYLE	MADE WITH	VERIFICATION	COMPILED
74 verified	Scientific	Grounded v0.5.1	Crossref · claims	4 Sep 2026

ABSTRACT

Replacing saturated fats with unsaturated oils lowers LDL cholesterol, while confidence in longer-term benefits depends on the clinical outcome and the replacement tested. Substitution trials offer some evidence of coronary benefit, while broader and omega-6-specific analyses leave important uncertainty about mortality. Recovered historical trials include adverse or null mortality findings; their mixed interventions and methodological limitations prevent a universal conclusion about current bottled oils. Reviews of common inflammatory markers are generally reassuring, but a newer controlled trial found changes in fatty-acid concentrations and stimulated blood-cell mediator production, making a blanket claim of biological neutrality untenable. Overall confidence is moderate for short-term lipid effects and lower for disease prevention from particular oils. Heating creates a separate chemical exposure, with very uncertain human clinical risk. A useful answer therefore specifies the oil, comparator, endpoint, and preparation conditions.

Figure 1 contrasts trial estimates with prospective biomarker associations; its two panels preserve the difference between these questions.

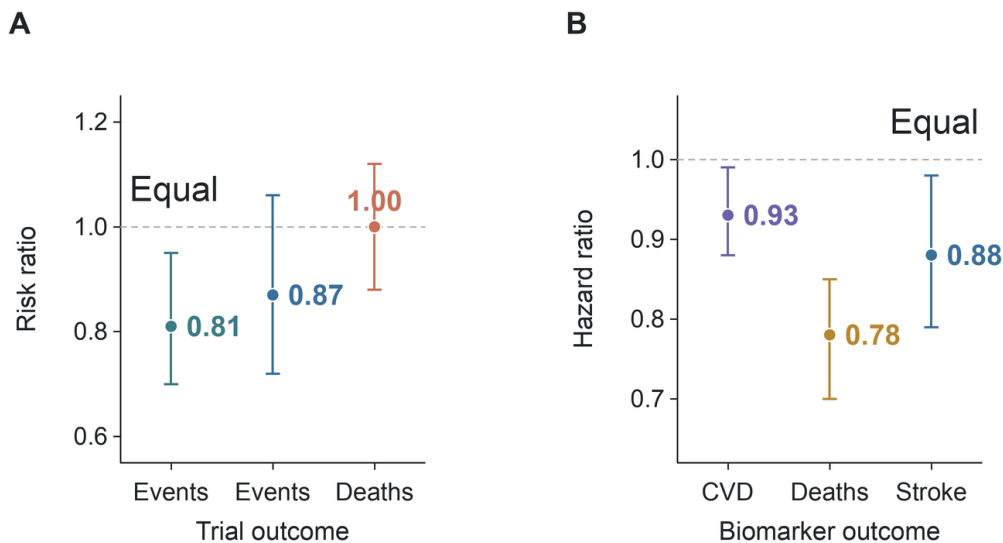


Figure 1. Trial and cohort estimates answer different questions. A substitution-trial meta-analysis estimated a coronary-event risk ratio of 0.81 (95% confidence interval 0.70–0.95).¹ A broader synthesis estimated 0.87 (0.72–1.06).² An omega-6 review estimated mortality at 1.00 (0.88–1.12).³ Prospective linoleic-acid biomarkers were associated with lower total cardiovascular disease.⁴ Vertical positions give relative estimates and whiskers their confidence intervals; horizontal categories identify reviews or outcomes. Trial points follow the source order above; CVD means cardiovascular disease.

01 Introduction

“Seed oils” is a popular category, not a standardized exposure. Analyses found linoleic-acid content spanning 1.6% to 79.0% of fatty acids across culinary oils.⁵ A Nordic scoping review notes that an oil’s effects on risk markers may vary with its polyphenols, phytosterols, and other minor com-

ponents.⁶ An origin label is consequently insufficient to specify a nutritional comparison.

The central question is not whether an oil came from a seed. It is which oil, used how, replaces which food, in whom, and for which outcome. This review separates five evidence layers: controlled changes in risk markers, randomized clinical events, prospective disease associations, mechanistic and inflammatory evidence, and degradation

during heating. These are distinct questions, so evidence for one cannot automatically answer the others.

The distinction between an intermediate marker and a clinical outcome is central. Lower LDL in a feeding comparison supports a claim about LDL; an observed association with mortality supports a claim about mortality within that study design. Neither result alone warrants a universal classification of an oil. The assessment below uses moderate confidence for short-term replacement effects on lipids and lower confidence for an effect of particular seed oils on disease. These judgments describe the evidence body, rather than the number of references supporting a sentence.

02 The category obscures substantial compositional differences

Vegetable oils vary in fatty-acid composition and minor components.⁶ An umbrella review reported different health benefits among vegetable oils, with lipid reductions especially for oils rich in monounsaturated and polyunsaturated fats.⁷ The Nordic scoping review reached a narrower conclusion: food-level oil syntheses were sparse and generally of low methodological quality.⁶ This is an important limitation because nutrient-level evidence cannot automatically

distinguish soybean oil from canola oil or an intact seed from oil in an ultra-processed food.

Linoleic and arachidonic acids have distinct physiological roles within omega-6 biology.⁸ Some mediators derived from the omega-6 fatty acid arachidonic acid are mainly pro-inflammatory, although ordinary dietary linoleic- and arachidonic-acid intake has shown little effect on inflammatory biomarkers.⁸ A label that combines high-linoleic sunflower oil, high-oleic sunflower oil, corn oil, canola oil, and partially hydrogenated historical margarines therefore creates a false single exposure. The differences become visible as soon as oils are compared directly.

03 Replacement determines the direction of the lipid effect

Controlled evidence consistently shows that replacing saturated fat with unsaturated fat lowers low-density lipoprotein cholesterol.⁹ A systematic review found convincing evidence that replacing saturated fat with polyunsaturated or monounsaturated fat lowers total and LDL cholesterol.⁹ A review focused on linoleic acid reached the same directional conclusion for lipid risk markers in healthy adults.¹⁰

The range in [Figure 2](#) retains butter as the reference throughout.

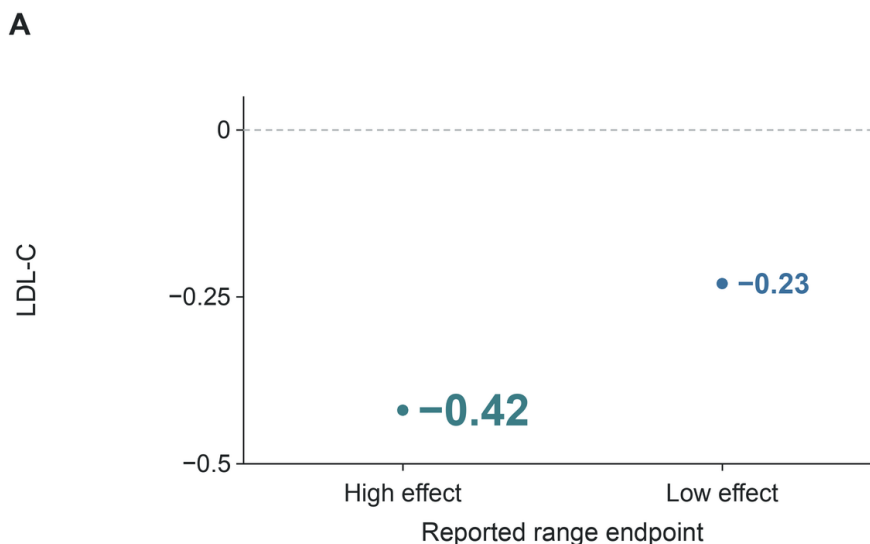


Figure 2. Oil replacement lowered LDL cholesterol relative to butter. A network meta-analysis of 54 trials reported an LDL-cholesterol range of -0.42 to -0.23 mmol/L per isocaloric replacement of 10% of dietary energy from butter with oils.¹¹ The vertical axis gives LDL differences in mmol/L; horizontal categories mark range endpoints, not named oils.

Head-to-head trials show why the comparator must be stated. In a crossover trial, corn oil produced a more favorable lipid profile than coconut oil over four weeks.¹² Replacing dairy fat with rapeseed oil lowered serum cholesterol, triglycerides, and LDL cholesterol in adults with hyperlipidaemia.¹³ A soybean-oil mayonnaise lowered total and LDL cholesterol relative to palm-olein mayonnaise, while leaving

oxidative-stress markers unchanged.¹⁴ Products low in saturated and trans fatty acids also improved lipoprotein cholesterol relative to more hydrogenated products.¹⁵ The cited endpoints support short-term metabolic comparisons; they do not establish fewer heart attacks.

The same oil can change rank when the reference changes. Coconut oil looked less adverse than butter but less favor-

able than cis-unsaturated oils in a current review.¹⁶ In a randomized three-way comparison, coconut oil was more comparable to olive oil with respect to LDL cholesterol.¹⁷ A systematic review of high-oleic oils found comparable lipid benefits when saturated or trans fats were replaced by high-oleic or high-polyunsaturated oils.¹⁸ Replacement is the causal question.

04 Coronary-event trials indicate modest benefit with uncertain mortality effects

Clinical-event syntheses are less uniform than lipid trials. A Cochrane review of increased polyunsaturated fat found little or no effect on all-cause mortality but a probable small reduction in coronary and cardiovascular events.² A separate meta-analysis of eight randomized trials, 13,614 participants, and 1,042 coronary events estimated a 19% event reduction when polyunsaturated fat replaced saturated fat (risk ratio 0.81, 95% confidence interval 0.70–0.95).¹ The estimate concerns substitution, not adding oil on top of an unchanged diet.

A review limited to omega-6 interventions judged the evidence less certain. It found little or no difference in all-cause mortality or cardiovascular events, with wide intervals for coronary outcomes.³ Ramsden’s analysis distinguished mixed omega-3/omega-6 interventions from omega-6-specific trials.¹⁹ Thus a linoleic-acid-specific question requires a narrower interpretation than a question about total polyunsaturated fat. The evidence supports a small possible coronary benefit more clearly than a survival advantage.

Because review overlap was not fully reconstructed, these syntheses were not treated as independent replications. Their different estimates warrant a conditional conclusion: there is some support for coronary benefit from particular replacement regimens, with lower confidence in attributing that benefit specifically to omega-6-rich oils. The uncertainty is substantive. A confidence interval spanning a favorable effect and little difference does not establish equivalence, and a favorable estimate from a broader intervention cannot remove uncertainty from a narrower one.

The clinical contrasts are kept separate in the following table.

SYNTHESIS	INCLUDED CONTRAST	CORONARY ESTIMATE	MORTALITY CONCLUSION	SOURCE
Broad PUFA review	Higher total PUFA	Coronary events probably slightly lower	Mortality little changed	²
Substitution trials	PUFA for saturated fat	RR 0.81 (95% interval 0.70–0.95)	Coronary endpoint	¹
Omega-6 review	Higher omega-6	Coronary effect uncertain	Mortality RR 1.00 (0.88–1.12)	³

Figure 3 shows how much uncertainty remains around each coronary estimate.

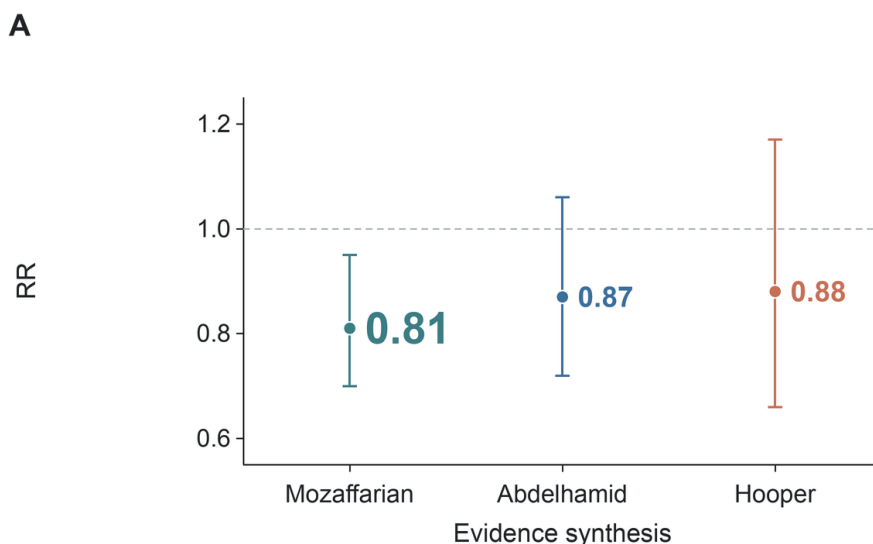


Figure 3. Randomized coronary-event estimates are modest and method-dependent. A substitution-focused synthesis reported a risk ratio of 0.81 (95% confidence interval 0.70–0.95).¹ A broader polyunsaturated-fat synthesis reported 0.87 (0.72–1.06).² An omega-6-specific synthesis reported 0.88 (0.66–1.17).³ Their visual alignment is not a new pooled estimate. Vertical position gives coronary risk ratios; horizontal categories identify syntheses. Whiskers show confidence intervals; one means no difference.

05 Recovered trials expose real uncertainty without proving a class-wide hazard

Recovered historical trials are the strongest challenge to a simple diet-heart story. In the Sydney Diet Heart Study, 458 men with recent coronary events were assigned to a safflower-oil intervention or usual care.²⁰ The intervention group had higher all-cause mortality (hazard ratio 1.62, 95% confidence interval 1.00–2.64) and cardiovascular mortality (1.70, 1.03–2.80).²⁰ A reanalysis separating omega-6-specific from mixed-polyunsaturated interventions similarly reported no benefit and a possible adverse direction for omega-6-specific trials.¹⁹ These findings cannot be dismissed merely because they are old.

The Minnesota Coronary Experiment provides a different caution. Replacing saturated fat with linoleic acid lowered serum cholesterol but did not lower coronary or all-cause mortality.²¹ The original Minnesota report likewise found no difference in cardiovascular events, cardiovascular deaths, or total mortality in the overall population.²² A stricter meta-analysis later argued that apparent benefit depended on including inadequately controlled trials.²³ Lower cholesterol and observed survival did not move together in these datasets.

However, these trials do not cleanly represent modern unhydrogenated oil use. The Sydney intervention included safflower-oil margarine as well as liquid oil.²⁰ The Minnesota diets used corn-oil polyunsaturated margarine against control margarines and shortenings.²¹ Reviews of partially hydrogenated oils show that expected coronary effects depend on both trans-fat content and the replacement oil.²⁴ Because these interventions changed more than a single cooking ingredient, their results constrain claims about the historical regimens rather than isolating the effect of a modern bottled oil. Historical null or adverse results are a legitimate boundary on certainty, but they are not a controlled test of every current seed oil.

That distinction also prevents an opposite overreach. The recovered analyses do not prove that lowering LDL is irrelevant; they show that a particular historical intervention failed to convert that change into longer survival. A broader review estimated about 10% lower coronary risk when 5% of energy from saturated fat was replaced by polyunsaturated fat, but found mixed evidence for several other cardiometabolic outcomes.²⁵ The defensible conclusion preserves both observations instead of forcing one to cancel the other.

06 Prospective biomarkers are favorable or neutral, but they remain observational

Prospective biomarker studies generally do not show cardiovascular harm from linoleic acid. In pooled international analyses, higher circulating or tissue linoleic acid was associated with lower total cardiovascular disease, cardiovascular mortality, and ischemic stroke.⁴ A later meta-analysis found that each standard-deviation increase in linoleic acid was associated with a 10% lower risk of fatal coronary disease, with no clear reduction in nonfatal disease.²⁶ In older adults, high circulating linoleic acid was associated with lower total and coronary mortality.²⁷ EPIC-Norfolk likewise found an inverse association between omega-6 phospholipid fatty acids and coronary disease.²⁸

Null results also matter. The Multi-Ethnic Study of Atherosclerosis found no significant association between omega-6 fatty acids and incident cardiovascular disease.²⁹ A Dutch cohort found that a 4–5 percentage-point energy difference in linoleic acid or carbohydrate intake was not associated with coronary incidence.³⁰ Adipose-tissue linoleic acid was not associated with myocardial infarction in a Danish case-cohort study.³¹ Across eleven cohorts, linoleic- and arachidonic-acid biomarkers were not associated with incident atrial fibrillation.³² Favorable and null estimates both argue against a large common hazard.

Biomarker associations require a causal interpretation before they can justify a dietary intervention. A Mendelian randomization analysis using genetic-association data from more than 114,000 UK Biobank participants did not support a protective role of circulating polyunsaturated-fatty-acid concentrations against cardiovascular disease and identified horizontal pleiotropy as a potential bias.³³ Cohorts constrain the plausible size of harm; they do not prove that adding a tablespoon of a particular oil prevents disease.

The assessment assigns low confidence to a causal cardiovascular benefit inferred from these biomarkers. The genetic analysis does not simply reverse the cohort findings: its own pleiotropy concern means that the genetic instruments may influence disease through other lipid-related routes.³³ Favorable observational estimates and uncertain genetic estimates therefore leave the dietary intervention question open. Confidence in a measured association can exceed confidence in the action someone might take on the basis of it.

Figure 4 keeps the cardiovascular and diabetes outcomes in separately identified categories.

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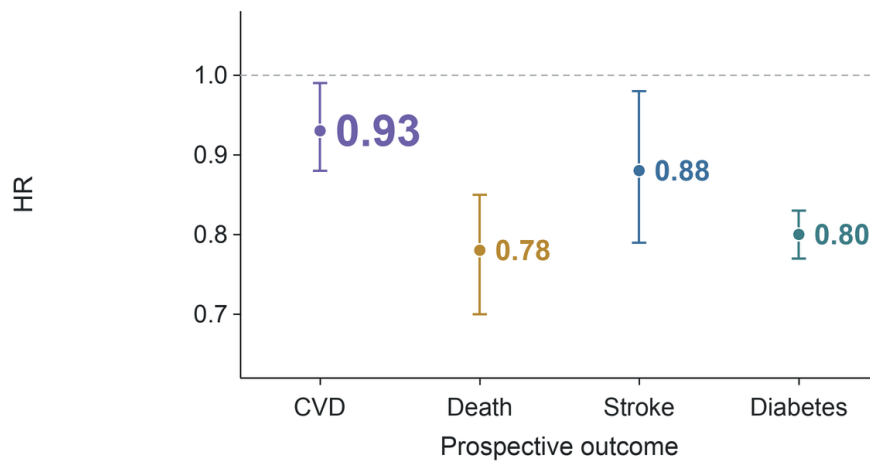


Figure 4. Circulating linoleic acid is favorable across selected prospective outcomes. Per interquintile-range increase in linoleic acid, pooled biomarker analyses reported HR 0.93 for total cardiovascular disease, 0.78 for cardiovascular mortality, and 0.88 for ischemic stroke.⁴ EPIC-InterAct reported HR 0.80 per standard-deviation increase in linoleic acid for its association with type 2 diabetes.³⁴ Vertical position gives hazard ratios; horizontal categories identify outcomes. Whiskers show confidence intervals. Exposure contrasts differ; associations are not oil-intervention effects.

07 Diabetes evidence differs between biomarkers and interventions

The diabetes literature repeats the cohort-trial gap. In a pooled analysis of 39,740 adults from 20 cohorts, higher linoleic-acid biomarkers were associated with lower incident type 2 diabetes.³⁵ EPIC-InterAct also found linoleic acid inversely associated with diabetes, while several less abundant omega-6 fatty acids had different directions.³⁴ These results again caution against treating an entire fatty-acid family as one exposure.

Randomized evidence is more restrained. A meta-analysis of omega-3, omega-6, and total polyunsaturated-fat trials found very uncertain effects of omega-6 on diabetes diagnosis and little or no effect on most measures of glucose metabolism.³⁶ By contrast, a meta-analysis of 102 controlled feeding trials found that replacing carbohydrate or saturated fat with polyunsaturated fat improved several measures of glucose-insulin regulation.³⁷ The trials measure different endpoints and durations: improved intermediate regulation does not guarantee fewer diagnoses, and a null diagnosis result does not erase short-term physiology.

08 Reviews found little evidence of higher common inflammatory markers

The claim that linoleic acid inevitably causes systemic inflammation has a direct randomized test. A systematic review of human interventions found virtually no evidence that adding linoleic acid increased common inflammatory markers in healthy adults.³⁸ A later meta-analysis also found no significant overall change in blood inflammatory markers when dietary linoleic acid increased.³⁹ A review

noted that high dietary or circulating linoleic acid was not associated with stronger in-vivo or ex-vivo inflammatory responses.⁴⁰

The assessment gives moderate confidence to the limited conclusion about the measured circulating markers. It does not equate a nonsignificant average change with absence of all inflammatory effects. The older reviews and the newer mechanistic experiment below address different endpoints, so the apparent disagreement should be resolved by naming what each actually measured.

09 Mechanistic plausibility cannot substitute for human outcomes

The net inflammatory interpretation of omega-6 biology is contested. A review of omega-6 inflammation describes both adverse and neutral or favorable signals.⁴¹ A mechanistic review emphasizes pro-inflammatory eicosanoids and associates a high omega-6-to-omega-3 ratio with chronic disease.⁴² A discussion of plasma-fatty-acid findings reported associations with lower pro-inflammatory and higher anti-inflammatory markers.⁴³ Mechanism supports questions; it does not settle population effects.

The dietary ratio remains contested for the same reason. A review argues that a lower omega-6-to-omega-3 ratio is broadly desirable.⁴⁴ A review of human studies found that few clinical studies tested the ratio directly and that their results conflicted.⁴⁵ A soybean-oil review reports that health agencies emphasize adequate intake of each fatty-acid family rather than a single ratio target.⁴⁶ A contrary review links seed-oil linoleic acid to oxidized metabolites and inflammatory pathways.⁴⁷ The ratio compresses two inde-

pendent intakes, multiple fatty acids, and a whole diet into one number.

The oxidized-linoleic-acid hypothesis is a more specific concern. A narrative review links linoleic-acid oxidation products to oxidized LDL and atherosclerotic lesions.⁴⁸ Toxicology reviews show that aldehydic lipid-oxidation products can form, enter food, and interact with biological systems.⁴⁹ Yet the latter review explicitly calls for nutritional and epidemiological trials connecting quantified exposure to chronic disease.⁴⁹ Plausibility is strongest for degraded oils under heat; evidence is weaker for ordinary unheated intake.

A newer randomized trial makes the endpoint distinction especially important. Sergeant and colleagues compared high- and low-linoleic-acid diets for twelve weeks, using safflower-oil variants while holding alpha-linolenic-acid exposure similar.⁵⁰ Of 80 randomized participants, 52 completed the intervention.⁵⁰ The high-linoleic-acid arm had lower plasma EPA, an omega-3 fatty acid, and higher ratios of arachidonic-acid-derived to EPA-derived products after laboratory stimulation of whole blood.⁵⁰ These findings make a blanket assertion of biological neutrality inappropriate.

The trial's clinical reach is narrower than its mechanistic result. The authors explicitly state that the study does not directly address clinical outcomes.⁵⁰ They also describe substantial dietary support, no common-diet run-in, and a reciprocal increase in oleic acid when linoleic acid was reduced.⁵⁰ The assessment therefore gives moderate confidence to the observed biochemical contrast and much less confidence to any prediction of disease from it. A stimulated blood assay is useful evidence about that response; it is not a cardiovascular event.

The practical research question is consequently more precise than whether omega-6 is inflammatory. Future trials could test whether these mediator changes persist and predict clinical outcomes under named replacement diets. That would connect the measured pathway to the outcome at issue without assuming that the pathway either proves harm or has no importance.

10 Named oils differ, but most head-to-head evidence is short term

Canola oil illustrates both the value and limitation of named-oil comparisons. A meta-analysis found lower LDL cholesterol, total cholesterol, and the LDL/HDL ratio with canola versus olive oil.⁵¹ In women with polycystic ovary syndrome, a ten-week trial reported lipid-profile improvements with canola compared with sunflower and olive oils.⁵² In patients referred for coronary angiography, canola and olive oil produced different inflammatory-marker patterns without clear overall lipid differences.⁵³ A high-oleic canola intervention altered LDL composition without increasing LDL proteoglycan binding.⁵⁴ These outcomes support biochemical comparisons in the studied populations; they do not establish durable clinical benefits.

Corn and soybean comparisons generally favor the less saturated option. Corn oil improved atherogenic lipids relative to extra-virgin olive oil in a controlled feeding trial.⁵⁵ Corn oil produced a more favorable plasma lipid profile than coconut oil in another trial.¹² Palm oil raised LDL cholesterol relative to vegetable oils lower in saturated fat.⁵⁶ Hydrogenating linoleic acid toward trans or stearic fatty acids raised LDL and lowered HDL relative to linoleic acid.⁵⁷ A palm-oil crossover trial found only modest changes when palm replaced common dietary fats.⁵⁸ Formulation and comparator explain much of the apparent conflict.

COMPARISON	DURATION	MAIN FINDING	SOURCE
Corn vs coconut	4 weeks	Better plasma lipid profile with corn	12
Soybean mayonnaise vs palm-olein mayonnaise	4 weeks	Lower total and LDL cholesterol with soybean mayonnaise	14
Canola vs olive	Trial meta-analysis	Lower LDL with canola	51

11 Food-level associations remain vulnerable to confounding

In a cohort of 521,120 older US adults, corn and canola oil were not associated with higher cardiometabolic mortality per tablespoon, while modeled replacement of butter or margarine with nonhydrogenated vegetable oils was associated with lower mortality.⁵⁹ A Chinese cross-sectional analysis associated vegetable/sesame-oil use with higher prevalent atherosclerotic cardiovascular disease than lard or other animal-fat use.⁶⁰ Its cross-sectional design cannot establish whether oil choice preceded disease. The contrary association deserves inclusion, but describing it as a prospective cohort would give it a temporal interpretation it does not have.

Broader dietary evidence explains why isolated-oil estimates vary. An etiologic synthesis found that estimated effects of individual dietary components were similar to quantified effects of dietary patterns on cardiovascular risk factors and clinical endpoints.⁶¹ A recent macronutrient meta-analysis found that total-fat associations changed when analyses were stratified by fat type.⁶² These food-level associations should consequently be interpreted with the same caution about the replacement and surrounding diet.

More recent prospective evidence points in the favorable direction without removing that limitation. In three US health-professional cohorts, the highest total plant-oil intake was associated with lower total mortality than the lowest intake, with a hazard ratio of 0.84 and a 95% confidence interval of 0.79–0.90.⁶³ The analysis used re-

peated dietary questionnaires and modeled replacements of butter with plant oils.⁶³ This is contemporary evidence for an association, not random assignment to the replacement. Its abstract supports the reported contrast; it cannot justify details absent from that accessible account.

The old and new cohort results do not need to be merged into a universal verdict. They strengthen the case for studying specific substitutions while leaving the estimated size of a causal benefit uncertain. That is why the assessment retains low certainty for disease prevention inferred from the food-level evidence, even where the reported association is precise.

12 High-temperature reuse creates a separate oxidation question

Heating changes the exposure. In a controlled experiment, oxidation products rose from 100°C to 200°C, and several aldehydes were highest in polyunsaturated soybean oil at 200°C.⁶⁴ Oxidative stability fell about 30% across the tested refined oils after twelve months of storage.⁶⁵ Rapeseed-oil experiments detected tertiary degradation products at advanced stages of lipid oxidation.⁶⁶ Heating conditions are therefore part of the exposure definition.

Figure 5 reports laboratory endpoints, preserving the boundary before any clinical interpretation.

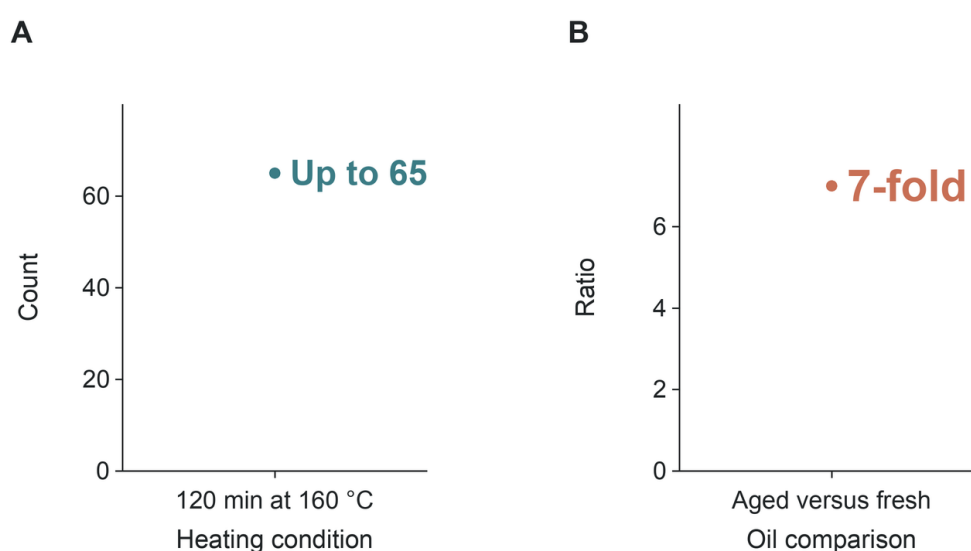


Figure 5. Extended heating changes the chemical exposure. Rapeseed oil incubated at 160°C for 120 minutes yielded up to 65 detected volatile compounds.⁶⁶ After 10 minutes, aged oil produced a 7-fold higher hexanal amount than non-aged oil.⁶⁶ Vertical axes give compound counts (A) and hexanal ratios (B); horizontal categories identify conditions.

Evidence for human disease remains less developed. Reviews identify oxidation products in abused frying oils as a plausible health concern.⁶⁷ A detailed review of aldehydic products calls for tolerable-intake values and direct epidemiological or nutritional studies.⁶⁸ An experimental study found tissue injury after rats consumed repeatedly heated oil.⁶⁹ That animal result supports hazard identification, not a quantitative estimate of risk from ordinary home cooking. The scientifically defensible boundary is to distinguish fresh or gently heated oil from repeatedly heated frying oil without pretending that human outcome data are complete.

The assessment separates moderate confidence in altered oil chemistry from very low confidence in the magnitude of clinical harm at ordinary human exposure. An oil sample producing a compound is a different endpoint from a person developing disease. The next useful evidence would measure both actual cooking exposure and subsequent

health, with enough detail to distinguish oil identity from heating conditions. A generic oil name cannot resolve that missing exposure definition.

13 Other disease endpoints remain uncertain

The cancer literature does not provide a single precise estimate for ordinary seed-oil exposure. A review concluded that high linoleic-acid intake was unlikely to substantially raise breast, colorectal, or prostate cancer risk, while acknowledging that a small increase could not be excluded.⁷⁰ A meta-analysis of 47 randomized trials found small point estimates with substantial uncertainty for cancer effects of total polyunsaturated fat, with sparse omega-6-specific evidence.⁷¹ A newer global cohort synthesis associated higher dietary or circulating omega-6 with lower overall cardiovascular, cancer, and mortality risks, but reported heterogeneity by cancer site and health status.⁷² The cancer conclusion remains limited because exposure definitions and sites differ.

Cognition and bowel disease also require their own evidence. A systematic review of randomized trials found that the effects of increasing omega-6 or total polyunsaturated fat on cognition were unclear.⁷³ Another randomized-trial synthesis found little support for modifying inflammatory bowel disease or long-term inflammatory status by increasing omega-3, omega-6, or total polyunsaturated fat.⁷⁴ These null findings reinforce the main pattern: fatty-acid composition can alter biomarkers, but broad chronic-disease claims require outcome-specific evidence.

14 Conclusion

The recurring constraint is specificity. An LDL result, a mortality association, a stimulated-cell response, and a heated-oil assay can all be valid observations without answering the same question. The most defensible clinical interpretation begins with the food being replaced and ends at the outcome actually measured. It does not convert the seed-oil label into either a universal hazard or an assurance that every formulation and use is beneficial.

The largest unresolved step is from those specified observations to long-term effects of named culinary oils. Trials of relevant replacement diets and cohorts with measured heating exposure would be more informative than another com-

parison that leaves both the comparator and preparation undefined. Until then, confidence should remain outcome-specific: firmer for short-term lipid changes, lower for disease effects attributed to particular oils, and very low for quantifying ordinary human risk from heating chemistry alone.

Scope and methods — Narrative review updated on 4 September 2026 using OpenAlex and PubMed, with ten search angles, review and primary-study searches, recent and contrary/null searches, and bidirectional citation chasing from six central sources. Human dietary interventions and observational findings were considered separately from laboratory and animal studies. This is not an exhaustive systematic review. Authenticated full texts and abstracts were used; full-text retrieval dates were preserved. Crossref screening found no integrity signal in the queried metadata for the included sources as of the check date; several publisher notice pages were inaccessible. Overlap between many reviews was not fully reconstructed and was not counted as independent corroboration. The certainty judgments are structured narrative assessments, not formal certainty ratings under the Grading of Recommendations Assessment, Development and Evaluation framework.

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107 CITED SENTENCES · 107 SOURCE CHECKS · 62 SUPPORTED AT FULL TEXT · 45 AT ABSTRACT · 0 PARTIAL · 0 CONTRADICTED

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